

ORF307: Optimization

Spring 2023 2nd Midterm Preparation Exercises

All the exercises appearing in the book chapters listed at the end of each lecture are great to prepare for the first midterm. Here is a list of representative exercises to guide your preparation:

1 Linear Programming: Foundations and Extensions [LP book]

Ch. 2 2.1 - 2.11, 2.16, 2.17, 2.18, 2.19

Ch. 3 3.2, 3.5, 3.6

Ch. 5 5.1, 5.14, 5.16

Ch. 6 6.6

Ch. 7 7.2

2 Linear optimization [LO book]

Ch. 2 2.8, 2.9, 2.10, 2.11, 2.13, 2.17

Ch. 3 3.2, 3.3, 3.4, 3.6, 3.10, 3.12, 3.18, 3.21

Ch. 4 4.1, 4.2, 4.4, 4.5, 4.10, 4.12, 4.13, 4.26, 4.27

Ch. 5 5.2, 5.3, 5.4, 5.6, 5.7, 5.10

Ch. 7 7.1, 7.2, 7.5, 7.21

3 Other exercises

1. *Bike production.* Company A has agreed to provide the following quantities of special bikes to Company B during the next 4 months.

- January: 150 units
- February: 160 units
- March: 225 units
- April: 180 units

Company A can produce a maximum of 160 bikes per month at a cost of \$35 per unit. Additional bikes can be purchased from Company C at a cost of \$50 per bike. Company A incurs an inventory holding cost of \$5 per month for each bike held in inventory. Company A starts from an empty inventory.

- (a) Formulate the problem that Company A is facing as a linear programming problem.

Hint. Given the inventory level h_t at time t , the quantity produced u_t , the units purchased p_t , and the quantity provided to another company d_t , the inventory dynamics can be modeled as $h_{t+1} = h_t + u_t + p_t - d_t$.

- (b) Verify that the following solution is optimal: produce 160 units in each month, keep 10 units in inventory from January to February and from February to March, keep no units in inventory from March to April, purchase no units from Company C in January and February, 55 units in March, and 20 units in April.
- (c) Company A is considering some preventative maintenance during one of the first three months. If maintenance is scheduled for January, the company can manufacture only 151 units (instead of 160); similarly, the maximum possible production if maintenance is scheduled for February to March is 153 and 155 units respectively. Which month out of the three would you recommend to have the maintenance? Explain why.

Solution.

(a) We can formulate the LP by creating the variables

- $u \in \mathbf{R}^4$: the production level in units for the four months
- $p \in \mathbf{R}^4$: the units purchased from Company C for the four months
- $h \in \mathbf{R}^4$: the inventory held from Jan to Feb, Feb to Mar, Mar to Apr, Apr to May

Then we have the LP

$$\begin{aligned} \text{minimize} \quad & 35\mathbf{1}^T u + 50\mathbf{1}^T p + 5\mathbf{1}^T h \\ \text{subject to} \quad & u_1 + p_1 = 150 + h_1 \\ & u_2 + p_2 + h_1 = 160 + h_2 \\ & u_3 + p_3 + h_2 = 225 + h_3 \\ & u_4 + p_4 + h_3 = 180 + h_4 \\ & u \leq 160 \\ & u, p, h \geq 0 \end{aligned}$$

(b) First transform the above problem into standard form (we need just 1 set of slack variables for the $u \leq 160$ constraint). We know that the point is optimal if the reduced cost vector is non-negative, i.e. $\bar{c} \geq 0$. By inspection, the basis is $B = \{1, 2, 3, 4, 7, 8, 9, 10\}$ and the vector of reduced costs

$$\bar{c} = (0, 0, 0, 0, 10, 5, 0, 0, 0, 0, 5, 55, 5, 10, 15, 15).$$

(c) If we have maintenance in January, then we see that we need to purchase 9 more units from Company C in March instead of producing 9 units in January and keeping it in inventory until March. This will induce an additional cost of $50 \cdot 9 - 9 \cdot (35 + 2 \cdot 5) = 45$. If we have maintenance in February, then we'll need to purchase 7 more units from Company C in March instead of producing 7 units in February and keeping it in inventory until March. This will induce an additional cost of $50 \cdot 7 - 7 \cdot (35 + 5) = 70$. If we have maintenance in March, then we'll need to purchase 5 more units from Company C in March instead of producing 5 units in March. This will induce an additional cost of $50 \cdot 5 - 5 \cdot (35) = 75$. So we see that the optimal choice is to have the maintenance in January.

2. *Duality.* Consider the linear programming problem

$$\begin{array}{ll}\text{minimize} & x_1 - x_2 \\ \text{subject to} & 2x_1 + 3x_2 - x_3 + x_4 \leq 0 \\ & 3x_1 + x_2 + 4x_3 - 2x_4 \geq 3 \\ & -x_1 - x_2 + 2x_3 + x_4 = 6 \\ & x_1 \leq 0 \\ & x_2, x_3 \geq 0.\end{array}$$

Write down the corresponding dual problem.

Solution. We can change the problem into standard form first. This involves flipping the sign of the greater or equal constraint, adding slack variables and replacing x_1 with its negative. We also need to replace x_4 with $x_4 = x_4^+ - x_4^-$, $x_4^+ \geq 0$, $x_4^- \geq 0$, obtaining

$$\begin{aligned} \text{minimize} \quad & -x_1^- - x_2 \\ \text{subject to} \quad & -2x_1^- + 3x_2 - x_3 + x_4^+ - x_4^- + s_1 = 0 \\ & 3x_1^- - x_2 - 4x_3 + 2x_4^+ - 2x_4^- + s_2 = -3 \\ & x_1^- - x_2 + 2x_3 + x_4^+ - x_4^- = 6 \\ & x_1^-, x_2, x_3, x_4^+, x_4^-, s_1, s_2 \geq 0. \end{aligned}$$

Writing in matrix notation

$$\begin{aligned} \text{minimize} \quad & \tilde{c}^T \tilde{x} \\ \text{subject to} \quad & \tilde{A} \tilde{x} = \tilde{b} \\ & \tilde{x} \geq 0, \end{aligned}$$

where $\tilde{c} = (-1, -1, 0, 0, 0, 0, 0)$, $\tilde{x} = (x_1^-, x_2, x_3, x_4^+, x_4^-, s_1, s_2)$, $\tilde{b} = (0, -3, 6)$ and

$$\tilde{A} = \begin{bmatrix} -2 & 3 & -1 & 1 & -1 & 1 & 0 \\ 3 & -1 & -4 & 2 & -2 & 0 & 1 \\ 1 & -1 & 2 & 1 & -1 & 0 & 0 \end{bmatrix}.$$

Then we can take the dual and directly write it in matrix form.

$$\begin{aligned} \text{maximize} \quad & -\tilde{b}^T y \\ \text{subject to} \quad & \tilde{A}^T y + \tilde{c} \geq 0. \end{aligned}$$

3. *Duality with symmetric matrices.* Let A be a symmetric square matrix (i.e., $A = A^T$). Consider the linear programming problem

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & Ax \geq c \\ & x \geq 0.\end{array}$$

Show that if x^* satisfies $Ax^* = c$ and $x^* \geq 0$, then x^* is an optimal solution.

Solution. After multiplying by -1 and changing sign of the inequality, the dual of the LP can be written as

$$\begin{aligned} & \text{maximize} && c^T z \\ & \text{subject to} && A^T z \leq c \\ & && z \geq 0. \end{aligned}$$

Now, we observe the feasible region of the primal and dual only intersect when $Ax = c$ and $x \geq 0$. Thus, if $x^* = Ac$ and $x^* \geq 0$, x^* is feasible for both the primal and dual. Since the objective values are the same for both problems as well, x^* is an optimal solution.

4. ∞ -norm and duality. Consider the ∞ -norm approximation problem

$$\text{minimize } \|Ax - b\|_\infty,$$

where $A \in \mathbf{R}^{m \times n}$ with $m \geq n$ and A has linearly independent columns. Let x_∞ the optimal solution (or one of the optimal solutions if there are multiple ones).

- (a) Write the problem above as an LP in inequality form.
- (b) Derive the dual LP.
- (c) You do not have to prove it, but it is possible to simplify the dual as follows

$$\begin{aligned} &\text{maximize } b^T z \\ &\text{subject to } A^T z = 0 \\ &\quad \|z\|_1 \leq 1. \end{aligned}$$

Now, consider the least-squares solution

$$x_{\text{ls}} = \text{argmin } \|Ax - b\| = (A^T A)^{-1} A^T b,$$

and denote the least squares residual as $r_{\text{ls}} = b - Ax_{\text{ls}}$. Assuming that $r_{\text{ls}} \neq 0$, show that

$$\hat{z} = -r_{\text{ls}} / \|r_{\text{ls}}\|_1, \quad \text{and} \quad \tilde{z} = r_{\text{ls}} / \|r_{\text{ls}}\|_1,$$

are feasible for the dual problem.

- (d) By duality, we know that $b^T \hat{z}$ and $b^T \tilde{z}$ are lower bounds for $\|Ax_\infty - b\|_\infty$. Which one is better?

Hint. From least-squares theory, we know that $A^T r_{\text{ls}} = 0$. Therefore, we have that $x_{\text{ls}}^T A^T r_{\text{ls}} = 0$.

Remark. This is useful when you need to approximately solve an extremely large ∞ -norm problem. Instead, you can solve the least-squares problem (which is much faster), and estimate the suboptimality in terms of ∞ -norm.

Solution.

(a) LP formulation

$$\begin{aligned} & \text{minimize} && t \\ & \text{subject to} && \begin{bmatrix} A & -\mathbf{1} \\ -A & -\mathbf{1} \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix} \leq \begin{bmatrix} b \\ -b \end{bmatrix}. \end{aligned}$$

(b) Dual problem

$$\begin{aligned} & \text{maximize} && -b^T u + b^T v \\ & \text{subject to} && A^T u - A^T v = 0 \\ & && \mathbf{1}^T u + \mathbf{1}^T v = 1 \\ & && u \geq 0, v \geq 0 \end{aligned}$$

(c) From the expression $x_{\text{ls}} = (A^T A)^{-1} A^T b$ we note that

$$A^T r_{\text{ls}} = A^T (b - A(A^T A)^{-1} A^T b) = A^T b - A^T b = 0.$$

Therefore $A^T \hat{z} = 0$ and $A^T \tilde{z} = 0$. Obviously we also have $\|\hat{z}\| = 1$ and $\|\tilde{z}\| = 1$, so $\hat{z}$ and $\tilde{z}$ are dual feasible.

(d) We have

$$b^T \hat{z} = \frac{-b^T r_{\text{ls}}}{\|r_{\text{ls}}\|_1} = \frac{(Ax_{\text{ls}} - b)^T r_{\text{ls}}}{\|r_{\text{ls}}\|_1} = -\frac{\|r_{\text{ls}}\|_2^2}{\|r_{\text{ls}}\|_1}$$

and, similarly,

$$b^T \tilde{z} = \frac{\|r_{\text{ls}}\|_2^2}{\|r_{\text{ls}}\|_1}.$$

Therefore, $b^T \tilde{z} > b^T \hat{z}$ and $\tilde{z}$ gives a better bound than $\hat{z}$.

5. *UpAndDown Inc.* Your company UpAndDown Inc., is a small airline that operates a single commuter plane in a “hopping flight” configuration. This means that the plane serves n cities (conveniently numbered 1 to n), and every day it visits them in a fixed sequence $\{1, 2, \dots, n\}$. The plane can pick up passengers at any node and drop them off at any node. The number of passengers that would like to go over arc j is given by b_j . The fare (per passenger) associated with this trip is f_j . The maximum number of passengers the plane can safely carry at a time is equal to C . Your task is to help the airline determine the optimal loading/unloading of passengers in order to maximize the total fare collected per trip, while never exceeding the plane capacity.

- (a) Give a formulation of this problem as a min-cost flow network optimization problem. Define clearly the nodes and arcs of your problem and what the costs and capacities of each arc are.

Hint. Define nodes for every city and every origin-destination pair.

- (b) How would you compute the sensitivity of the optimal cost with respect to a small change in the number of passengers that want to fly over connection j , i.e., the number b_j ?

Solution.

- (a) The new graph has $n + \binom{n}{2}$ nodes corresponding to the cities $i = 1 \dots n$ and the origin-destination pairs $j = 1 \dots \binom{n}{2}$. There are new arcs from each origin-destination node j to the cities that correspond to that arcs origin and destination. The first arc has infinite capacity and cost $-f_j$. The second arc has infinite capacity and zero cost. Finally, there are arcs from city i to city $i + 1$, with no cost but finite capacity C . The influx to the origin-destination node j is b_j , and the outflow from city i is $\sum_{j \in A} b_j$ where A is the set of all origin-destination trips that end in city i .
- (b) The sensitivity of the cost from the dual solution (node potentials).

6. *Column-stochastic matrices.* An $n \times n$ matrix is called *column-stochastic* if all its entries are non-negative and the sum of the entries in each column is equal to 1 (i.e., the columns represent probabilities). In other words, P is column-stochastic if

$$P_{ij} \geq 0, \quad i, j = 1, \dots, n, \quad \text{equivalently} \quad P \geq 0,$$

and

$$\sum_{i=1}^n P_{ij} = 1, \quad j = 1, \dots, n, \quad \text{equivalently} \quad P^T \mathbf{1} = \mathbf{1}.$$

- (a) Show that if P is column-stochastic, then for any vector x we have

$$(P^T x)_i \leq x_{\max},$$

where $x_{\max} = \max_i x_i$ is the largest entry of x . In other words, we have

$$P^T x \leq x_{\max} \mathbf{1}.$$

- (b) Assume that P is column-stochastic. Show using LP duality that there exists a $y \in \mathbf{R}^n$ such that

$$Py = y, \quad y \geq 0, \quad \mathbf{1}^T y = 1.$$

Hint. Write an LP and its dual, and analyze the feasibility/unboundedness using the result in (a).

Solution.

(a)

$$P^T x = \begin{bmatrix} P_{11} & \dots & P_{n1} \\ \vdots & & \vdots \\ P_{1n} & \dots & P_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^n P_{i1} x_i \\ \vdots \\ \sum_{i=1}^n P_{in} x_i \end{bmatrix} \leq \begin{bmatrix} \sum_{i=1}^n P_{i1} x_{\max} \\ \vdots \\ \sum_{i=1}^n P_{in} x_{\max} \end{bmatrix} = \begin{bmatrix} x_{\max} \\ \vdots \\ x_{\max} \end{bmatrix}$$

Therefore, we have $(P^T x)_i \leq x_{\max}$.

(b) Consider the following LP problem

$$\begin{aligned} & \text{minimize} && -\mathbf{1}^T x \\ & \text{subject to} && (P - I)x = 0 \\ & && x \geq 0 \end{aligned}$$

This LP is always feasible (as $x = 0$ is a feasible solution). Its dual is written as

$$\begin{aligned} & \text{maximize} && 0 \\ & \text{subject to} && (P - I)^T y - \mathbf{1} \geq 0 \end{aligned}$$

Let $j = \arg \max_{i=1}^n y_i$. From (a) we know $(P^T y)_j \leq y_{\max}$. Then we have $((P - I)^T y - \mathbf{1})_j = (P^T y)_j - y_j - 1 \leq y_{\max} - y_j - 1 = -1$. Therefore, the dual LP is infeasible, and as a consequence the primal is unbounded. Thus, there exists x such that $(P - I)x = 0, x \geq 0, -\mathbf{1}^T x = -1$, which is equivalent to $Px = x, x \geq 0, \mathbf{1}^T x = 1$.

7. Consider the problem

$$\begin{array}{ll}\text{minimize} & -2x_1 - x_2 \\ \text{subject to} & x_1 - x_2 \leq 2 \\ & x_1 + x_2 \leq 6 \\ & x_1, x_2 \geq 0\end{array}$$

- (a) Convert the problem into standard form and construct a basic feasible solution at which $(x_1, x_2) = (0, 0)$
- (b) Carry out one iteration of the simplex method. Report the values for $\bar{c}$, θ , d , and x (including any additional variables you added to make it into standard form) in each iteration. You can use any programming language or calculator to help you with the computations.
- (c) Compute the next $\bar{c}$. What would you do if you wanted to complete the simplex method?

Solution.

1. In standard form

$$\begin{array}{ll}\text{minimize} & -2x_1 - x_2 \\ \text{subject to} & x_1 - x_2 + s_1 = 2 \\ & x_1 + x_2 + s_2 = 6 \\ & x_1, x_2, s_1, s_2 \geq 0\end{array}$$

An initial basic feasible solution is $(0, 0, 2, 6)$.

2. $\bar{c} = c - A^T p$ where $A_B^T p = c_B$. Here we have $B = (3, 4)$, $c_B = (0, 0)$ and

$$A_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

So $p = (0, 0)$ and $\bar{c} = (-2, -1, 0, 0)$. So we could have either x_1 or x_2 enter the basis; we take x_1 by Bland's rule. The search direction d solves the system $A_B d_B = -A_j$. Here, $A_j = (1, 1)$ so $d = (1, 0, -1, -1)$. We see that $\theta = 2$ to force s_1 to leave the basis. The new x is $x = (2, 0, 0, 4)$.

3. Now we compute the next round of reduced costs. $B = (1, 4)$, $c_B = (-2, 0)$ and

$$A_B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

So $p = (-2, 0)$ and $\bar{c} = (0, -3, 2, 0)$. Since $\bar{c}_2 < 0$, we know that we need to do at least one more simplex iteration.